

AI Readiness Training Program — POC Repository

Program Overview

This repository contains the complete documentation, user stories, implementation guides, and test specifications for the **AI Readiness Training Program** — a structured 5-phase upskilling initiative for ICD NGAs of all practices (Java / Microsoft / DTS / QET)

Program Goal: Equip bench NGAs with practical AI engineering skills — from building full-stack applications with GitHub Copilot, to building RAG pipelines, context-aware agents, MCP servers, and multi-agent systems — making them deployment-ready for future project allocations.

How to Use This Repository

1. The mentor will assign a POC from the approved repository list. Once assigned, the POC is final, and no changes or reassignments will be permitted
2. **Read the `overview.md`** in your assigned POC folder to understand the full scope
3. **Start Phase 1** by reading `phase1-fullstack-crud.md` — follow it step by step
4. **After completing each phase**, run the test suite in `tests/phase[N]-test-spec.md` and record your pass/fail counts
5. **Submit results** to your mentor at the end of each phase (5-day phase window)
6. **Progress linearly** — each phase builds directly on the previous one

Important: You implement the same POC across all 5 phases. You can not ask to switch POCs mid-program.

Phase Overview

Phase	Name	Duration	Score Weight	What You Build
Phase 1	Full Stack CRUD with Copilot	5 days	15%	End-to-end full stack app (REST API + React UI + SQLite) built via GitHub Copilot vibe coding
Phase 2	RAG Application	5 days	20%	LangChain RAG pipeline (ChromaDB + Gemini 2.0 Flash) that answers questions about your app's user manual
Phase 3	Context Engineering & Tool Integration	5 days	20%	Expand RAG to call your Phase 1 REST API as tools; ReAct agent with context-engineered prompts
Phase 4	MCP Server & Chat Interface	5 days	25%	Convert REST API to MCP server; build a Streamlit chat interface connected via LangChain

Phase	Name	Duration	Score Weight	What You Build
Phase 5	Multi-Agent with LangGraph	5 days	20%	LangGraph StateGraph with 4 specialized agents orchestrated by a Supervisor pattern

Total Program Duration: ~25 working days (5 weeks)

Technology Stack

Layer	Technology
Backend (choose one)	Python 3.11 + FastAPI + SQLAlchemy + Alembic
	.Net 8 + ASP.NET Core Web API + Entity Framework Core
	Java 21 + Spring Boot 3 + Spring Data JPA
Frontend (React mandatory + one more)	React 18 + Vite + Axios
	Angular 17 / Blazor / Thymeleaf / Streamlit
Database	SQLite
LLM	Google Gemini 2.0 Flash (Google AI Studio free tier)
Vector Database	ChromaDB
AI Orchestration (P2–P4)	LangChain + langchain-google-genai
AI Orchestration (P5)	LangGraph
MCP Framework (P4)	fastmcp (Python)
Observability	LangSmith + structlog/Serilog/SLF4J + OpenTelemetry
Testing	pytest + httpx / xUnit / JUnit 5
Vibe Coding	GitHub Copilot (pre-provisioned)

Scoring & Performance Tiers

Phase Weights

Phase	Weight	Max Score	Pass Threshold (70%)
Phase 1	15%	15 points	10.5 points
Phase 2	20%	20 points	14 points
Phase 3	20%	20 points	14 points
Phase 4	25%	25 points	17.5 points
Phase 5	20%	20 points	14 points

Phase	Weight	Max Score	Pass Threshold (70%)
Total	100%	100 points	70 points

Performance Tiers

Tier	Criteria	Label
 Elite Performer	Cleared all 5 phases (≥70% in each)	POCs are presented to Account Delivery Heads to demonstrate competencies and enhance opportunities for project allocations
 Emerging Contributor	Cleared exactly 4 phases	Only a select group of shortlisted NGAs will be invited to present the POC to the Account Delivery Heads (ADHs)
 Foundational Builder	Cleared 3 or fewer phases	Will receive additional mentorship and be assigned targeted training programs to enhance their skills and meet the expected standards

A phase is "cleared" when ≥70% of that phase's automated test cases pass.

Repository Structure

```
d:/ICD_NGA_AI Readiness_Program_POCs/
├─ README.md                ← You are here
├─ TECH_STACK_REFERENCE.md  ← Environment setup for all 3 stacks
├─ SCORING_RUBRIC.md        ← Test categories, weights, evaluation
├─ guide
│   └─ OBSERVABILITY_GUIDE.md ← LangSmith + OpenTelemetry setup guide
├─ Assigned POC/
│   ├── overview.md
│   ├── phase1-fullstack-crud.md
│   ├── phase2-rag-application.md
│   ├── phase3-context-engineering.md
│   ├── phase4-mcp-chat-interface.md
│   ├── phase5-multi-agent.md
│   └─ tests/
│       ├── phase1-test-spec.md
│       ├── phase2-test-spec.md
│       ├── phase3-test-spec.md
│       ├── phase4-test-spec.md
│       └─ phase5-test-spec.md
└─ ...

## Quick Start

### Step 1: POC Assigned
Once the POC is assigned, review all the relevant files thoroughly and reach out to your mentor for clarification on any uncertainties
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Step 2: Set Up Your Environment

Read `TECH\_STACK\_REFERENCE.md` and complete the setup for your chosen stack (Python / .Net / Java). Set up Google AI Studio API key, LangSmith account, and GitHub Copilot.

Step 3: Begin Phase 1

Open the assigned POC folder and read `overview.md`, then `phase1-fullstack-crud.md`. Follow the 5-day schedule and implementation guide.

Step 4: Run Tests After Each Phase

After completing implementation, run the test suite from `tests/phase[N]-test-spec.md`. Record your pass/fail counts and submit to your mentor.

Step 5: Progress to Next Phase

Each phase builds on the previous. Do not skip ahead – Phase 2 requires Phase 1's running application.

Mentoring Schedule

Day	Session	Duration
Day 1 of each phase	Concept introduction + Q&A	1 hour
Day 2-3	Implementation check-in	1 hour
Day 4	Troubleshooting + code review	1 hour
Day 5	Test run + phase wrap-up	1 hour

> Mentor contact: ****Basavaraj Bidanoor / L B Rakesh Kumar****

> Mentoring channel: ****Microsoft Teams****

Reference Documents

Document	Purpose
`TECH_STACK_REFERENCE.md`	Step-by-step environment setup for Python, .Net, Java, Google AI Studio, LangSmith, ChromaDB, GitHub Copilot
`SCORING_RUBRIC.md`	Detailed scoring breakdown, test case categories, evaluation criteria, AI quality metrics
`OBSERVABILITY_GUIDE.md`	LangSmith tracing setup, OpenTelemetry span guide, structured logging standards for all 5 phases